

Drone Flight Controller, Motor Fault, and SBUS Debugging

This English version is formatted for graduate application review. It preserves the engineering purpose, design decisions, validation evidence, and individual contribution signals of the original course document while presenting the material in concise English.

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Project	Quadcopter Ball-Transfer Drone
Course	Practice of Mechanical Engineering
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Primary Role	Signal-chain and integration debugger
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Executive Summary

This note documents troubleshooting work around flight-controller setup, motor behavior, and SBUS signal communication. It captures the kind of integration work that decides whether a physical robot can be tested at all.

The engineering challenge was to avoid guessing. A flight failure could come from wiring, channel mapping, controller configuration, motor output, calibration, signal protocol, or vibration. The work focused on isolating these causes systematically.

Applicant Contribution Signals

- Worked through command path, receiver/SBUS behavior, flight-controller configuration, and motor response.
- Used step-by-step checks to identify whether faults belonged to hardware, communication, or configuration.
- Strengthened the final drone by making the electronics stack more understandable to the whole team.

Debugging Strategy

Trace the signal path from controller input to receiver output, flight-controller interpretation, ESC command, and motor response.

Verify one interface at a time so that later flight tests do not hide basic communication failures.

System Impact

Signal-chain confidence allowed the team to focus final testing on flight behavior and mission strategy instead of repeatedly reopening basic electronics uncertainty.

Technical Decision Table

Constraint	Decision	Why It Mattered	Skill Demonstrated
Motor response fault	Checked power, controller output, and ESC/motor path	Prevented blind tuning around a hardware issue	Electrical debugging
SBUS uncertainty	Traced receiver and channel behavior	Improved confidence in command input	Communication protocols
Flight-controller setup	Verified configuration before flight tests	Reduced final-test surprises	Embedded integration

How to Read This Evidence

This document is intended to help a reviewer quickly understand the engineering content of the original report in English. The original PDF remains available in the project archive as the source artifact with its original figures, tables, screenshots, and course formatting.